Example 4.6

Find (i) $\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)$ (ii) $\cos^{-1}\left(\cos\left(-\frac{\pi}{3}\right)\right)$ (iii) $\cos^{-1}\left(\cos\left(\frac{7\pi}{6}\right)\right)$

Solution

It is known that $\cos^{-1} x : [-1, 1] \to [0, \pi]$ is given by

$\cos^{-1} x = y$ if and only if $x = \cos y$ for $-1 \le x \le 1$ and $0 \le y \le \pi$.

Thus, we have

(i) $\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right) = \frac{3\pi}{4}$, since $\frac{3\pi}{4} \in [0, \pi]$ and $\cos\frac{3\pi}{4} = \cos\left(\pi - \frac{\pi}{4}\right) = -\cos\frac{\pi}{4} = -\frac{1}{\sqrt{2}}$.

(ii) $\cos^{-1}\left(\cos\left(-\frac{\pi}{3}\right)\right) = \cos^{-1}\left(\cos\left(\frac{\pi}{3}\right)\right) = \frac{\pi}{3}$, since $-\frac{\pi}{3} \notin [0, \pi]$, but $\frac{\pi}{3} \in [0, \pi]$.

(iii) $\cos^{-1}\left(\cos\left(\frac{7\pi}{6}\right)\right) = \frac{5\pi}{6}$, since $\cos\left(\frac{7\pi}{6}\right) = \cos\left(\pi + \frac{\pi}{6}\right) = -\frac{\sqrt{3}}{2} = \cos\left(\frac{5\pi}{6}\right)$ and $\frac{5\pi}{6} \in [0, \pi]$.

Example 4.7

Find the domain of $\cos^{-1}\left(\frac{2 + \sin x}{3}\right)$.

Solution

By definition, the domain of $y = \cos^{-1} x$ is $-1 \le x \le 1$ or $|x| \le 1$. This leads to

$-1 \le \frac{2 + \sin x}{3} \le 1$ which is same as $-3 \le 2 + \sin x \le 3$.

So, $-5 \le \sin x \le 1$ reduces to $-1 \le \sin x \le 1$, which gives

$-\sin^{-1}(1) \le x \le \sin^{-1}(1)$ or $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$.

Thus, the domain of $\cos^{-1}\left(\frac{2 + \sin x}{3}\right)$ is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

EXERCISE 4.2

1. Find all values of x such that

   (i) $-6\pi \le x \le 6\pi$ and $\cos x = 0$ (ii) $-5\pi \le x \le 5\pi$ and $\cos x = -1$.

2. State the reason for $\cos^{-1}\left[\cos\left(-\frac{\pi}{6}\right)\right] \ne -\frac{\pi}{6}$.

3. Is $\cos^{-1}(-x) = \pi - \cos^{-1}(x)$ true? Justify your answer.

4. Find the principal value of $\cos^{-1}\left(\frac{1}{2}\right)$.

5. Find the value of

   (i) $2\cos^{-1}\left(\frac{1}{2}\right) + \sin^{-1}\left(\frac{1}{2}\right)$ (ii) $\cos^{-1}\left(\frac{1}{2}\right) + \sin^{-1}(-1)$

   (iii) $\cos^{-1}\left(\cos\frac{\pi}{7}\cos\frac{\pi}{17} - \sin\frac{\pi}{7}\sin\frac{\pi}{17}\right)$.

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